Candidate Name: ________________________ Class: _____________

PIONEER JUNIOR COLLEGE
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JC2 PRELIMINARY EXAMINATION
Higher 2

MATHEMATICS **9740/02**
Paper 2
17 September 2014
3 hours

Additional Materials: Cover page
Answer papers
List of Formulae (MF15)

READ THESE INSTRUCTIONS FIRST

Write your full name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.
Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.
You are expected to use an approved graphing calculator.
Unsupported answers from a graphing calculator are allowed unless a question specifically states otherwise.
Where unsupported answers from a graphing calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.
You are reminded of the need for clear presentation in your answers.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

Section A: Pure Mathematics [40 marks]

1 **(i)** Express $\dfrac{6n+2}{(n+1)(n+2)(n+3)}$ in partial fractions. [2]

(ii) Hence find $\displaystyle\sum_{r=2}^{n}\left(\frac{3r+1}{(r+1)(r+2)(r+3)}\right)$ in the form of $c-\mathrm{f}(n)$, where c is a constant to be determined. [3]

(iii) Deduce $\displaystyle\sum_{k=2}^{n}\left(\frac{3k-2}{k(k+1)(k+2)}\right)$. [2]

2 The complex number z satisfies the equation

$$|6i-8-2z|=|6-8i|.$$

(i) Sketch an Argand diagram to illustrate this equation. [3]

(ii) Find the smallest and largest possible values of $\arg(z+4i-4)$. [4]

3 **(i)** Find $\displaystyle\int x^2\mathrm{e}^x\ \mathrm{d}x$. [3]

(ii) Hence, find the exact volume of the solid of revolution formed when the region bounded the curve $y=x\mathrm{e}^{\frac{1}{2}x}$, the lines $y=2\mathrm{e}$, $x=3$ and the x-axis is rotated through 4 right angles about the x-axis. [4]

4 **(i)** Given that $\sin\sqrt{y}=x$, prove that

$$\left(1-x^2\right)\frac{\mathrm{d}^2y}{\mathrm{d}x^2}=x\frac{\mathrm{d}y}{\mathrm{d}x}+2.$$ [2]

(ii) By further differentiation, find the series expansion for y in ascending powers of x, up to and including the term in x^4. [4]

Denote the answer to part **(ii)** by Y.

(iii) Find, for $-1\le x\le 1$, the range of values of x for which the value of Y differ from y by less than 0.05. [3]

5 The equation of a line l_1 is given by

$$\mathbf{r}=\begin{pmatrix}-3\\10\\-5\end{pmatrix}+\lambda\begin{pmatrix}2\\-4\\1\end{pmatrix},\ \lambda\in\mathbb{R}.$$

(i) The perpendicular to l_1 from a point $A(-1,-2,6)$ meets l_1 at the point N. Find the position vector of N. [4]

Another line l_2 passes through A and the point $(-3,10,-5)$.

(ii) Find a cartesian equation of the line which is a reflection of l_2 in l_1. [3]

(iii) The plane p is perpendicular to l_2. Find the angle between p and l_1. [3]

Section B: Statistics [60 marks]

6 Events A and B are such that P(A) = $k+\frac{1}{6}$, P(B) = $k+\frac{1}{12}$ and P($A\cup B$) = $\frac{1}{2}$.

Given A and B are independent, find the value of k. [3]

7 In a certain company, there are 3 female employees in the Accounts Department, 4 female and 2 male employees in the Marketing Department, 2 female and 5 male employees in the Production Department and 2 female and 2 male employees in the Research Department. At a particular meeting, a committee of 3 members is to be chosen at random from all employees in the company to lead a special project. Find the probability that

(i) exactly one member is from the Accounts Department, [2]

(ii) all three members are from the different departments, [2]

(iii) at least two male members are from the Production Department. [2]

8 A company develops a new type of fibre and claims that its mean breaking strength exceeds 50 Newtons. A random sample of 10 fibres is selected and their breaking strengths, in Newtons, are as follows:

48.73, 52.79, 51.31, 51.05, 49.23, 50.65, 52.01, 50.52, 49.43, 50.59.

(i) Stating a necessary assumption, carry out an appropriate test of the company's claim at the 4% significance level. [5]

Explain, in the context of the question, the meaning of 'at the 4% significance level'. [1]

(ii) It is given instead that the breaking strength has a variance of 1.5. In a one-tail test, the company's claim is accepted. Find an inequality satisfied by the significance level of the test. [3]

9 A female doctor wishes to test a new treatment for a certain genetic disease. On a particular day, she asks n (where n is fixed) patients who come to her clinic suffering from this disease if they are willing to take part in a trial. Based on past experiences, she finds that the probability of such a patient willing to take part is 0.3. The number of patients who are willing to take part is the random variable R.

(i) State, in the context of this question, two assumptions needed to model R by a binomial distribution. [2]

(ii) Explain why one of the assumptions stated in part **(i)** may not hold in this context. [1]

Assume now that these assumptions do in fact hold.

(iii) Given that $n = 8$, find the probability that R is at least 2. [1]

(iv) If the chance of at least 2 patients willing to take part in this trial is more than 90%, find the least possible integer value of n. [3]

(v) Explain why the sample of size n obtained is not random.

In fact, the new treatment will be conducted for patients from different age groups. Explain how a quota sample of 50 might be obtained in this context.

[3]

10 **(i)** Explain why it is advisable to plot a scatter diagram before interpreting a linear correlation coefficient calculated for a sample drawn from a bivariate distribution. [1]

The table gives the values of six observations of bivariate data, x and y.

x	25	30	35	40	45	50
y	0.067	0.125	1.131	1.000	3.330	7.627

(ii) Calculate the product moment correlation coefficient between x and y, and explain whether your answer suggests that a linear model is appropriate.

[2]

(iii) Draw a scatter diagram for the data. [1]

One of the values of y appears to be incorrect.

(iv) Indicate the corresponding point on your diagram by labelling it P, and explain why the scatter diagram for the remaining points may be consistent with a model of the form $\ln y = a + bx$. [2]

(v) Omitting P, calculate the least squares estimate of a and b for the model $\ln y = a + bx$. [2]

(vi) Assume that the value of x at P is correct. Estimate the value of y for this value of x. [1]

(vii) In fact, the variable y represents the percentage of pregnant women of age x years giving birth to a child with Down's Syndrome. It is required to estimate the age of a woman for which $y = 1.131$. Explain why neither the regression line of x on y nor the regression line of x on $\ln y$ should be used. [1]

11 The random variable X denotes the lifespan of a battery in hours, which may be taken to have a normal distribution with mean μ and standard deviation σ. Given that $\mathrm{P}(X < 800) = 0.023$ and $\mathrm{P}(X > 1100) = 0.281$, find the values of μ and σ, correct to 2 decimal places. [4]

(i) A random sample of 200 batteries is taken. Using suitable approximation, find the probability that at least 190 batteries in this sample have lifespan of at least 800 hours. [3]

(ii) A random sample of n batteries is taken. Find the largest value of n if the probability of the average lifespan of the sample is at most 1000 hours exceeds 0.1. [3]

12 A manufacturer produces two types of machines, A and B. He estimates that breakdowns in A occur at an average rate of 1 per 48 hours of use while breakdowns in B occur at an average rate of 1 per 60 hours of use.

It is assumed that the breakdowns for the machines follow Poisson distributions.

A company buys one unit of A and one unit of B, and uses them throughout the day daily.

(i) Find the probability that, in a randomly chosen week, the total number of breakdowns in the machines is less than 4. [2]

(ii) In a randomly chosen week, the total number of breakdown in machines is less than 4. Find the probability that there are more breakdowns in A than in B. [4]

(iii) Using a suitable approximation, calculate the probability that in a 100-day period, the number of breakdown in A differs from twice that in B by more than 5. [5]

(iv) State the assumption used in parts **(i)** to **(iii)**. [1]